Group Food Decision Web Application — Product Specifications Document

Executive Summary

The Group Food Decision Web Application is a mobile-first, session-based platform designed to eliminate the friction of collective dining decisions. By aggregating individual preferences, budget constraints, and real-time location data, the application delivers a single, consensus-driven restaurant recommendation in under two minutes. Built entirely on open-source technologies, this MVP prioritizes speed, simplicity, and conflict-free group coordination without requiring user accounts.

Problem Statement

Groups of people—friends, coworkers, students—frequently experience decision paralysis when choosing where to eat together. Contributing factors include:

- **Divergent budgets**: Some members may prefer affordable options while others are willing to spend more
- **Conflicting food preferences**: Dietary restrictions, cuisine preferences, and taste variations create friction
- **Information asymmetry**: Lack of awareness of nearby affordable options leads to repetitive choices or prolonged debates
- **Social dynamics**: Dominant voices override quieter members, leading to dissatisfaction

These factors result in wasted time (often 10–30 minutes), interpersonal conflict, and suboptimal dining experiences.

Objectives

Objective	Metric
Reduce group decision time	< 2 minutes from session start to final recommendation
Eliminate decision conflict	Single output, no voting disputes

Maximize group satisfaction	≥ 80% acceptance rate of recommendations
Ensure accessibility	Mobile-first, no account required
Maintain open-source compliance	Zero proprietary dependencies

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Target Users

Primary Personas

Persona	Description	Key Needs
College Friend Group	3–6 students deciding on dinner	Low budget, variety, speed
Office Lunch Crew	4–8 coworkers with limited break time	Proximity, quick service, dietary options
Casual Social Group	2–5 friends meeting spontaneously	Discovery, consensus, minimal effort

User Characteristics

- Age: 18–40
- Tech-savvy, smartphone-first
- Low patience for complex onboarding
- Value speed over feature richness

Scope Definition

In-Scope (MVP)

- Session creation with shareable link/code
- Individual preference input (cuisine, budget, dietary restrictions)
- Location-based restaurant discovery via OpenStreetMap
- Algorithmic recommendation engine

- Single final recommendation display
- Session expiration (24-hour TTL)

Out-of-Scope (Future Iterations)

- User accounts and authentication
- Reservation integration
- Restaurant reviews/ratings aggregation
- Payment splitting
- Chat/messaging within session
- Native mobile applications
- Multi-language support

User Flow

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│                  USER FLOW                       │
└──────────────────────────────────────────────────────────────
──────────────────────────────────────┘
[1] HOST CREATES SESSION
│
▼
[2] HOST SETS LOCATION (auto-detect or manual)
│
▼
[3] HOST RECEIVES SHAREABLE LINK/CODE
│
▼
[4] HOST SHARES LINK WITH GROUP
│
▼
[5] PARTICIPANTS JOIN VIA LINK
│
▼
[6] EACH PARTICIPANT ENTERS:
• Display name
• Budget range ($ / $$ / $$$)
• Cuisine preferences (multi-select)
```

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• Dietary restrictions (optional)
  |
  ▼
[7] HOST SEES REAL-TIME PARTICIPANT LIST
  |
  ▼
[8] HOST TRIGGERS "FIND RESTAURANT"
  |
  ▼
[9] ALGORITHM PROCESSES:
• Location radius
• Budget intersection
• Cuisine overlap
• Dietary filters
  |
  ▼
[10] SINGLE RECOMMENDATION DISPLAYED
• Restaurant name
• Address + map
• Price range
• Cuisine type
• Distance
  |
  ▼
[11] SESSION COMPLETE
• Option to regenerate (1 retry)
• Option to start new session
```

Functional Requirements

FR-01: Session Management

ID	Requirement	Priority
FR-01.1	System shall generate unique 6-character alphanumeric session codes	Must
FR-01.2	System shall create shareable URLs containing session code	Must
FR-01.3	Sessions shall expire after 24 hours of inactivity	Must

FR-01.4	Host shall be identified by session creation, stored in browser session	Must
FR-01.5	Maximum 10 participants per session	Should

FR-02: Location Services

ID	Requirement	Priority
FR-02.1	System shall request browser geolocation permission	Must
FR-02.2	System shall allow manual address entry as fallback	Must
FR-02.3	System shall geocode addresses using Nominatim (OSM)	Must
FR-02.4	Default search radius shall be 2 km, adjustable to 1/5/10 km	Should

FR-03: Preference Collection

ID	Requirement	Priority
FR-03.1	Participants shall enter display name (2–20 characters)	Must
FR-03.2	Participants shall select budget tier: $ (0–10), $$ (10–25), $$$ (25+)	Must
FR-03.3	Participants shall select 1–5 cuisine preferences from predefined list	Must
FR-03.4	Participants may select dietary restrictions (vegetarian, vegan, halal, gluten-free)	Should
FR-03.5	Preferences shall be editable until recommendation is triggered	Should

FR-04: Recommendation Engine

ID	Requirement	Priority
FR-04.1	Engine shall query restaurants within specified radius	Must
FR-04.2	Engine shall filter by lowest common budget tier	Must

FR-04.3	Engine shall rank by cuisine preference overlap score	Must
FR-04.4	Engine shall exclude restaurants violating any dietary restriction	Must
FR-04.5	Engine shall return single top-ranked result	Must
FR-04.6	If no match, system shall display "No restaurants match criteria" with retry option	Must

FR-05: Result Display

ID	Requirement	Priority
FR-05.1	Result shall display restaurant name, address, cuisine, price tier	Must
FR-05.2	Result shall display interactive map via Leaflet	Must
FR-05.3	Result shall include "Get Directions" link (opens external maps)	Should
FR-05.4	Host may trigger one re-roll for alternative recommendation	Should

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Non-Functional Requirements

NFR-01: Performance

ID	Requirement	Target
NFR-01.1	Page load time	< 2 seconds (3G network)
NFR-01.2	Recommendation generation	< 3 seconds
NFR-01.3	API response time	< 500ms (p95)
NFR-01.4	Concurrent sessions supported	1,000

NFR-02: Availability & Reliability

ID	Requirement	Target

NFR-02.1	Uptime	99.5%
NFR-02.2	Data persistence	Session data retained for 24 hours
NFR-02.3	Graceful degradation	Functional without JavaScript for basic info

NFR-03: Security

ID	Requirement	Target
NFR-03.1	HTTPS enforcement	All traffic encrypted
NFR-03.2	Session codes	Cryptographically random, non-sequential
NFR-03.3	Input validation	All user inputs sanitized
NFR-03.4	Rate limiting	100 requests/minute per IP

NFR-04: Usability

ID	Requirement	Target
NFR-04.1	Mobile responsiveness	Full functionality on 320px+ screens
NFR-04.2	Accessibility	WCAG 2.1 AA compliance
NFR-04.3	Onboarding	Zero-instruction usability

NFR-05: Maintainability

ID	Requirement	Target
NFR-05.1	Code coverage	≥ 70% unit test coverage
NFR-05.2	Documentation	README, API docs, inline comments
NFR-05.3	Containerization	Full Docker deployment

Success Metrics

Metric	Definition	Target (MVP)

Decision Time	Time from session creation to recommendation display	< 2 minutes
Session Completion Rate	% of sessions reaching recommendation	≥ 75%
Participant Join Rate	Avg participants per session	≥ 3
Recommendation Acceptance	% of sessions not using re-roll	≥ 80%
System Uptime	Monthly availability	≥ 99.5%
Page Load Performance	Lighthouse mobile score	≥ 85

Assumptions & Dependencies

Assumptions

- Users have smartphones with modern browsers (Chrome, Safari, Firefox)
- Users are willing to share location data
- OpenStreetMap data is sufficiently accurate for target regions
- Restaurant data can be sourced/seeded from open datasets (Overpass API, OpenStreetMap POIs)

Dependencies

- Nominatim API availability for geocoding
- Overpass API for restaurant POI data
- Leaflet.js library maintenance
- PostgreSQL 14+ compatibility

Risks & Mitigations

Risk	Impact	Probability	Mitigation
Insufficient restaurant data in OSM	High	Medium	Seed database with curated local data; allow community

			contributions post-MVP
Nominatim rate limiting	Medium	Low	Implement caching; self-host Nominatim if needed
Low adoption due to link-sharing friction	Medium	Medium	Generate QR codes for in-person sharing
Algorithm produces poor recommendations	High	Medium	Implement feedback mechanism; iterate on scoring weights

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Glossary

Term	Definition
Session	Temporary group instance for decision-making
Host	User who creates and controls the session
Participant	User who joins an existing session
Preference Overlap	Intersection of cuisine preferences across participants
Budget Tier	Price category ($, $$, $$$)
TTL	Time-to-live; session expiration period

Document Version: 1.0

Status: MVP Specification Complete

Next: Deliverable 2 — UI/UX Design Specification